

Project AETHER

Advanced Electronic Tactical Hybrid Emergency Reconnaissance UAV

Abstract—Project AETHER is a modular autonomous unmanned aerial system developed for the 2026 Student Unmanned Aerial Systems (SUAS) competition. The platform is designed around disaster response, environmental awareness, and autonomous mission execution. The UAV integrates thermal reconnaissance, autonomous navigation, real-time telemetry, and embedded edge processing into a lightweight and adaptable aerial platform capable of supporting emergency response operations.

The system architecture combines a high-efficiency multirotor airframe with onboard sensing and communication subsystems including thermal imaging, mmWave radar, GPS, long-range telemetry, and autonomous flight control. A distributed embedded architecture centered around an ESP32 processing unit and flight controller enables low-latency sensor fusion, real-time environmental monitoring, and resilient communication during mission operations.

The design philosophy emphasizes modularity, operational reliability, and rapid deployment in disaster environments such as flood zones, fire incidents, and search-and-rescue operations. Extensive subsystem-level validation, communication testing, and integration planning were performed to ensure compatibility between sensing, telemetry, and autonomous navigation systems. The final design provides a scalable foundation for future expansion into AI-assisted target recognition, obstacle avoidance, and fully autonomous mission planning.

I. COMPETITION STRATEGY

The SUAS 2026 competition emphasizes autonomous mission execution, environmental awareness, endurance, and operational resilience. To maximize mission effectiveness, Team Karasuno adopted a systems-engineering-based approach that prioritizes subsystem integration, modularity, and rapid field deployment.

Instead of optimizing only for endurance or payload delivery, the team focused on creating a balanced reconnaissance platform capable of:

- Autonomous waypoint navigation
- Real-time thermal reconnaissance
- Environmental monitoring
- Long-range telemetry
- Emergency beacon deployment
- Future AI-assisted object detection

The project was divided into five major engineering divisions.

The overall strategy aligns closely with the 2026 storm-response mission profile by emphasizing situational awareness, low-latency communication, and reliable autonomous operation.

II. SYSTEM DESIGN STRATEGY

Project AETHER was designed using a modular systems engineering methodology. Each subsystem was independently

TABLE I
ENGINEERING DIVISIONS

Subsystem	Primary Responsibility
Airframe & Propulsion	Structural design and flight stability
Embedded Systems	ESP32 integration and sensor management
Communication & Telemetry	LoRa, WebSocket, and ExpressLRS systems
Reconnaissance & Sensors	Thermal imaging and radar sensing
Autonomous Systems	Navigation and future AI-based autonomy

developed and validated before integration into the complete UAV architecture. This reduced debugging complexity and enabled rapid iteration during development.

The UAV architecture was designed around four primary engineering objectives:

- Reliable autonomous flight
- Real-time environmental sensing
- Robust communication systems
- Lightweight and modular construction

A quadcopter configuration was selected due to its:

- Mechanical simplicity
- High maneuverability
- Stable hovering capability
- Efficient power-to-weight ratio

The airframe integrates carbon-fiber structural elements and lightweight electronics mounting systems to minimize weight while maintaining structural rigidity.

III. AIRFRAME AND PROPULSION SYSTEM

The propulsion system was selected to provide stable flight performance while supporting multiple reconnaissance sensors and communication modules.

TABLE II
HARDWARE CONFIGURATION

Component	Selection
Flight Controller	GEPRC F722
Motors	VELOX V2812 BLDC Motors
ESC	50A 4-in-1 ESC
Battery	6S Li-Ion Battery Pack

The propulsion architecture prioritizes endurance and stable low-speed flight for reconnaissance missions. A 6S Li-Ion configuration was selected due to its improved energy density compared to conventional LiPo batteries.

The propulsion layout was optimized to:

- Reduce vibration transmission
- Improve thermal dissipation
- Simplify maintenance
- Maintain center-of-gravity stability

IV. EMBEDDED ELECTRONICS AND PROCESSING

A distributed embedded architecture was implemented to separate flight-critical tasks from sensor-processing tasks.

A. ESP32 Processing Unit

The ESP32 microcontroller serves as the onboard sensor-processing hub and communication bridge. Responsibilities include:

- Thermal camera interfacing
- Sensor fusion
- Beacon control
- WebSocket communication
- LoRa transmission
- Real-time telemetry handling

The ESP32 communicates with peripheral systems using:

- UART
- PWM
- I²C
- WebSocket protocols

This architecture reduces computational load on the flight controller and improves subsystem scalability.

V. SENSOR AND RECONNAISSANCE SYSTEMS

The reconnaissance subsystem was developed to support disaster monitoring and emergency response operations.

TABLE III
INTEGRATED SENSORS

Sensor	Purpose
MLX90640 Thermal Camera	Fire and heat-source detection
GPS Module	Positioning and navigation
mmWave Radar	Environmental awareness
Visual Camera	Real-time video feed

The MLX90640 thermal camera enables real-time thermal mapping for:

- Fire hotspot detection
- Survivor identification
- Disaster-zone monitoring

The mmWave radar subsystem enhances environmental awareness under low-visibility conditions such as smoke, rain, or fog.

VI. COMMUNICATION AND TELEMETRY

Reliable communication is critical during emergency-response missions. Project AETHER integrates multiple communication systems for redundancy and operational resilience.

The telemetry dashboard was developed using JavaScript-based visualization tools and supports:

TABLE IV
COMMUNICATION ARCHITECTURE

System	Purpose
ExpressLRS	Long-range remote control
LoRa	Emergency beacon transmission
WebSocket	Real-time dashboard telemetry

- Live thermal feed
- Sensor monitoring
- UAV status visualization
- GPS tracking

The LoRa subsystem enables low-bandwidth long-range emergency beacon transmission even in degraded network environments.

VII. SOFTWARE ARCHITECTURE

The software stack was designed around modularity and rapid prototyping.

TABLE V
SOFTWARE STACK

Software	Role
Betaflight	Flight control firmware
Arduino IDE	Embedded firmware development
JavaScript Dashboard	Telemetry visualization

The architecture supports future integration with:

- ROS2
- MAVLink
- AI-based object recognition
- Autonomous path planning

VIII. AUTONOMOUS MISSION CAPABILITY

Project AETHER is being developed toward semi-autonomous and fully autonomous operation.

Current autonomous capabilities include:

- GPS waypoint navigation
- Autonomous stabilization
- Telemetry-based monitoring
- Embedded sensor coordination

Future upgrades planned for SUAS deployment include:

- AI-assisted object recognition
- Autonomous obstacle avoidance
- Dynamic mission replanning
- Thermal target classification

The architecture is designed to support NVIDIA Jetson integration for future onboard machine learning acceleration.

IX. TESTING STRATEGY

Testing followed a progressive integration methodology where individual components were validated prior to subsystem integration.

A. Component-Level Testing

- Motor functionality validation
- ESC calibration
- Sensor communication testing
- Battery voltage verification

B. Communication Testing

- LoRa transmission range testing
- ExpressLRS signal reliability
- WebSocket telemetry validation

C. Thermal Reconnaissance Testing

- Thermal image acquisition
- Temperature mapping verification
- Real-time dashboard visualization

D. Flight Testing

- Manual stabilization flights
- Hover stability validation
- Telemetry monitoring
- Sensor synchronization verification

X. RISK MANAGEMENT

Several technical and operational risks were identified during development.

TABLE VI
RISK MANAGEMENT

Risk	Mitigation
Electrical failure	Redundant voltage regulation
Communication loss	Long-range backup telemetry
Overheating	Passive airflow cooling
Sensor failure	Modular replacement architecture
Flight instability	PID tuning and vibration damping

Operational safety procedures include:

- Pre-flight inspection checklists
- Battery health verification
- Failsafe configuration
- Emergency manual takeover capability

XI. EXPECTED PERFORMANCE

Project AETHER is expected to provide:

- Stable autonomous reconnaissance flight
- Real-time thermal monitoring
- Long-range telemetry transmission
- Disaster-area environmental awareness
- Emergency beacon deployment capability

The modular design also enables future expansion into:

- AI-based detection systems
- Autonomous rescue coordination
- Multi-UAV cooperative operations

XII. CONCLUSION

Project AETHER demonstrates the integration of embedded systems, autonomous flight control, and disaster-response sensing into a unified unmanned aerial platform. The system was designed specifically to align with the operational philosophy of SUAS 2026 by emphasizing resilience, reconnaissance capability, modularity, and autonomous functionality.

Through the integration of thermal imaging, telemetry systems, and embedded sensor processing, the UAV provides a scalable architecture capable of supporting real-world emergency-response missions. Future development will focus on advanced autonomy, AI-assisted perception, and enhanced mission adaptability.

REFERENCES

- [1] Betaflight Flight Controller Documentation.
- [2] Espressif ESP32 Technical Reference Manual.
- [3] MLX90640 Thermal Camera Datasheet.
- [4] ExpressLRS Documentation.
- [5] LoRa Communication Protocol Documentation.